

Master Cheat Sheet: OpenWrt 5-Port Dumb Switch Conversion

Project: Transforming an Old TP-Link TL-WR841N v13 into a Dedicated Layer-2 Ethernet Switch
Primary Router: Tenda TX2 Pro V2 (1000 Mbps Gigabit DHCP Server)
Secondary Device: TP-Link TL-WR841N v13 (100 Mbps Fast Ethernet Switch)

1. Network Architecture & Hardware Realities

Device Role & Subnet Breakdown

Device	Assigned Role	IP Address	Port Speed
Tenda TX2 Pro V2	Primary Gateway, Firewall, DHCP & DNS Server	192.168.0.1	1000 Mbps (Gigabit)
TP-Link TL-WR841N v13	Layer-2 Wired Dumb Switch (Wi-Fi Disabled)	192.168.0.250 (Static)	100 Mbps (Fast Ethernet)

Optimal Device Connection Map

- **Connect to Tenda TX2 Pro (1000 Mbps):** Pi Server, PC/Desktop (Requires maximum local transfer speeds).
- **Connect to TP-Link Switch (100 Mbps):** CCTV Camera, Home Laptop, general smart home devices (Consumes <15 Mbps).
- **Total Available LAN Ports: 6 Usable Ports** across both devices (4 downstream ports on TP-Link + 2 remaining ports on Tenda).





2. A-to-Z Complete Build Procedure

Step 1: Download the Correct OpenWrt Firmware

- **Hardware Target:** MediaTek MT7628AN (ramips/mt76x8).
- **File to Download:** Latest Stable Release (not SNAPSHOT) -> ...-squashfs-tftp-recovery.bin (used for initial factory flashing on v13).

Step 2: Flash the Router

- **Method A (Standard Stock Web UI):**
 1. Connect PC to Yellow LAN Port 1 on the TP-Link.
 2. Open 192.168.0.1 (or 192.168.1.1) in your browser.
 3. Go to **System Tools** -> **Firmware Upgrade**, select the downloaded .bin file, and upgrade.
- **Method B (TFTP Recovery Mode — if Stock UI throws Error 4503):**
 1. Rename downloaded firmware to exactly: **tp_recovery.bin**.
 2. Place file in a clean folder (e.g., C:\tftp).
 3. Set PC IPv4 address statically to **192.168.0.66** (Subnet: 255.255.255.0).
 4. Run **Tftpd64** as Administrator; set Current Directory to C:\tftp and Server Interface to 192.168.0.66.
 5. Power off router -> Hold **RESET** button -> Power on while holding RESET -> Hold for 10 seconds until file transfer finishes -> Release and wait 3–5 minutes.

Step 3: Disable 2.4 GHz Wi-Fi Radio

1. Access OpenWrt LuCI at http://192.168.1.1 (default OpenWrt IP after flash).
2. Go to **Network** -> **Wireless**.
3. Click **Disable** on the radio0 (2.4 GHz) wireless network to eliminate radio interference with the primary Tenda router.

Step 4: Delete Internet Routing (WAN Interfaces)

1. Go to **Network** -> **Interfaces**.
2. Click **Delete** next to **WAN** and **WAN6** (a Layer-2 switch does not route internet traffic).

Step 5: Convert Blue WAN Port to LAN Port 5

1. Go to **Network** -> **Switch** (switch0 / rt305x-esw).
2. Locate **VLAN 2** (the WAN VLAN) at the bottom of the table -> click **Delete**.
3. Locate **VLAN 1** (the LAN VLAN) -> change the dropdown for the **WAN port** from off to

untagged.

4. Click **Save & Apply** (All 5 ports: LAN 1–4 + WAN are now bridged into one physical switch).

Step 6: Set Static Management IP & Disable DHCP

1. Go to **Network -> Interfaces** -> click **Edit** next to **LAN**.
2. Under the **DHCP Server** tab (bottom): Check **Ignore interface** (stops OpenWrt from handing out IPs).
3. Under **General Settings**:
 - **IPv4 address**: Set to **192.168.0.250** (an unused IP outside Tenda's DHCP range).
 - **IPv4 netmask**: 255.255.255.0
 - **IPv4 gateway & Custom DNS**: 192.168.0.1 (Tenda's IP).
4. **CRITICAL SAVE STEP**: Click the **small down-arrow (▼)** next to **Save & Apply** -> select **Apply unchecked** (prevents OpenWrt's 90-second rollback timer from reverting your IP change).

Step 7: Final Hardware Interconnect

1. Set PC Ethernet adapter IPv4 back to **Obtain an IP address automatically (DHCP)**.
2. Plug the Ethernet cable from the **Tenda TX2 Pro V2** into **ANY** of the 5 ports on the TP-Link WR841N.
3. Your switch is online.

3. Maintenance, Management & Recovery

How to Access the Switch Admin GUI Later

- Open your browser from any device connected to your home network and go to: **http://192.168.0.250**.
- *Note*: You do not need to change your PC's IP address to access it; the Tenda router routes local traffic to .250 automatically.

How to Upgrade OpenWrt Firmware Later

1. Download the latest **...-sysupgrade.bin** image for TL-WR841N v13.
2. Navigate to **http://192.168.0.250** -> go to **System -> Backup / Flash Firmware**.
3. Under **Flash new firmware image**, upload the sysupgrade.bin file and keep **Keep settings** checked.

The 10-Second Emergency Hardware Reset

If you ever lock yourself out of the web interface or misconfigure the switch:

1. Keep the TP-Link router powered ON.
2. Press and hold the **RESET button** on the back for **10 seconds** until all LEDs flash rapidly, then release.
3. Wait **2 minutes**. OpenWrt is cleanly restored to factory default (192.168.1.1 with DHCP

enabled).

4. Problem-Solving Ledger (What Went Wrong & How to Solve It)

Error / Symptom	Root Cause	Exact Solution
Error Code: 4503 (<i>Stock UI Upgrade Rejected</i>)	TP-Link stock firmware restricts third-party firmware signatures or rejects long filenames.	Rename file to <code>tp_recovery.bin</code> and flash via TFTP Recovery Mode (PC static IP 192.168.0.66).
"Configuration changes have been rolled back!" (<i>90-Second Warning</i>)	Changing OpenWrt's IP disconnects your browser. If OpenWrt doesn't receive a confirmation ping within 90 seconds, it reverts changes to prevent lockouts.	Always click the ▼ down-arrow next to <i>Save & Apply</i> and choose Apply unchecked when changing LAN IP or DHCP settings.
ERR_CONNECTION_TIME_OUT on 192.168.0.3	Tenda cached an old DHCP lease (.0.3) for the OpenWrt WAN port. OpenWrt's firewall blocks HTTP GUI access from WAN by default.	Delete WAN interfaces in LuCI, use Yellow LAN ports for management, and navigate to the static LAN IP (192.168.0.250).
IP Address Conflict on 192.168.0.2	Another access point on the home network was already using .0.2, causing packets to collide.	Standardized the TP-Link switch management address to 192.168.0.250 (high IP, no conflicts).
Web GUI Unreachable, but Internet / Speedtest is 100% Perfect	OpenWrt reverted its GUI management IP to 192.168.1.1 after a rollback timer, while devices are on the 192.168.0.X subnet.	Core Rule: Ethernet switches operate at Layer 2 (MAC addresses) . IP addresses are only for human GUI access. An unmanaged switch does not need a GUI or IP address to switch packets at 100 Mbps flawlessly.

5. Summary Check-off (Repository Quick-Verify)

- [x] **Wi-Fi Radio Disabled:** Zero wireless frequency clash with Tenda TX2 Pro V2.
- [x] **DHCP Server Disabled (Ignore interface):** Tenda TX2 Pro V2 handles 100% of IP leases.
- [x] **WAN Port Untagged on VLAN 1:** 5 total usable Layer-2 wired switch ports.
- [x] **Static Management IP Assigned:** Pinned to 192.168.0.250 via Apply unchecked.
- [x] **Hardware Performance Verified:** 100 Mbps symmetrical throughput confirmed via Speedtest.